

The nonvanishing lemma for ribbon shape weights is false, and the true statement is a commuting-words theorem

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Abstract

Let R_e^W add an e -ribbon to a partition with a weight depending on the whole shape of the ribbon. The classification of the commuting pairs $[R_e^W, R_f^W] = 0$ in [1] rested on one unproved lemma: that no weight value may vanish. **That lemma is false.** For every $e < f$ with $\gcd(e, f) \geq 3$ there are commuting pairs in which almost every weight value is zero — at $(e, f) = (3, 6)$ the operator R_3^W that adds *only* the L-tromino $(2, 1)$, and nothing else, commutes with an explicit R_6^W supported on two ribbon shapes out of 32.

The repair is not a patch but a better theorem. Write $\sigma \otimes \rho$ for the shape weight of rank $e + f$ obtained by concatenating the occupancy words of σ and ρ across a free letter. Then the *entire* one-bead sector of the commutator says exactly

$$\bar{\sigma} = \sigma \otimes \rho = \rho \otimes \sigma,$$

that is, σ and ρ **commute** under $\otimes$; and the Lyndon–Schützenberger phenomenon holds in this graded semigroup: two commuting weights are, up to scalars, powers of a common weight τ of rank $d = \gcd(e, f)$. The proof is a four-line Euclidean descent and uses *no* nonvanishing hypothesis whatever. Everything then follows:

- the weight values vanish nowhere iff τ does;
- for $d \leq 2$ the rank- d two-bead relation forces τ nowhere zero, so Theorem 1.4 of [1] is **unconditionally true** there — in particular Corollary 1.5 (the coprime headline, and the decisive test at $(1, 3)$) is now a theorem, with the $\gcd(e, f) = 1$ case needing no two-bead input at all;
- for $d \geq 3$ it is **false** as stated, and the missing solutions are exactly the lifts of the rank- d weights with zeros.

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1 What this paper does

The claim it refutes

[1, Lemma 7.3] (“lem:nonvanish”) asserts: if $[R_e^W, R_f^{\bar{W}}] = 0$ and $W \not\equiv 0 \not\equiv \bar{W}$ over an integral domain, then no value of W or $\bar{W}$ is zero. It was proved there for $e \leq 2$ and left as gap (G1) for $e \geq 3$, with the remark that it is “the only gap in the classification”. The brief for the present session asked for a proof at $e \geq 3$.

It is false. The smallest counterexample is $(e, f) = (3, 6)$ and it is small enough to state in full (Theorem 6.7): let

$$W(\text{the ribbon } (2, 1)) = -1, \quad W = 0 \text{ on } (3) \text{ and } (1, 1, 1),$$

$$\bar{W}((2, 3, 1)) = +1, \quad \bar{W}((2, 1, 2, 1)) = -1, \quad \bar{W} = 0 \text{ on the other 30 shapes of size 6.}$$

Then $[R_3^W, R_6^{\bar{W}}] = 0$. Both operators are nonzero; indeed R_3^W is the operator that adds an L-tromino, and only an L-tromino.

The theorem that replaces it

Convention 1.1 (concatenation of shape weights). A *shape weight of rank $m \geq 1$* over a commutative ring F is a function $\sigma : \{0, 1\}^{m-1} \rightarrow F$. For shape weights σ_1, σ_2 of ranks m_1, m_2 define $\sigma_1 \otimes \sigma_2$, of rank $m_1 + m_2$, by

$$(\sigma_1 \otimes \sigma_2)(y) := \sigma_1(y_{[1, m_1-1]}) \sigma_2(y_{[m_1+1, m_1+m_2-1]}), \quad y \in \{0, 1\}^{m_1+m_2-1}.$$

The letter y_{m_1} is ignored. This is associative but not commutative; write $\tau^{\otimes k}$ for the k -fold power.

Theorem 1.2 (structure; §4). *Let σ_1, σ_2 be nonzero shape weights of ranks m_1, m_2 over a field F with $\sigma_1 \otimes \sigma_2 = \sigma_2 \otimes \sigma_1$, and put $d = \gcd(m_1, m_2)$. Then there are a nonzero shape weight τ of rank d and $c_1, c_2 \in F^\times$ with $\sigma_i = c_i \tau^{\otimes(m_i/d)}$.*

This is the Lyndon–Schützenberger theorem (“two words commute iff they are powers of a common word”) for the semigroup of Convention 1.1, and it is proved by the same Euclidean descent. Its relevance is Proposition 3.1: *the one-bead sector of the ribbon commutator is precisely the statement $\bar{\sigma} = \sigma \otimes \rho = \rho \otimes \sigma$* . So the gcd in [1, Thm. 1.4] is not an accident of the answer; it is the gcd of the commuting-words theorem.

Theorem 1.3 (classification; §6). *Let $1 \leq e < f$, $d = \gcd(e, f)$, K an integral domain with fraction field F , and let $W, \bar{W}$ be shape weights of ranks e, f over K with $W \not\equiv 0 \not\equiv \bar{W}$. Then $[R_e^W, R_f^{\bar{W}}] = 0$ if and only if there are a nonzero shape weight τ of rank d over F and $c, \bar{c} \in F^\times$ with*

$$\sigma = c\tau^{\otimes(e/d)}, \quad \bar{\sigma} = \bar{c}\tau^{\otimes(f/d)},$$

and σ satisfies the two-bead relation (3) of rank e (vacuous if $e = 1$). Here $\sigma(u) = (-1)^{|u|}W(u)$ as in [1, Conv. 2.1].

Corollary 1.4 (the corrected nonvanishing statement; §6). *In the situation of Theorem 1.3, W and $\bar{W}$ vanish nowhere if and only if τ does. Consequently [1, Lemma 7.3] is*

$$\text{true for } \gcd(e, f) \leq 2, \quad \text{false for } \gcd(e, f) \geq 3.$$

So the headline of [1] survives and is upgraded: for $\gcd(e, f) = 1$ the argument becomes unconditional, and — unexpectedly — needs no two-bead input at all (Corollary 6.2). What fails is the $\gcd \geq 3$ half of the classification, which was the half that was new.

Why the old route could not have worked

The abandoned strategy was to prove that the zero set of σ is empty by propagating zeros through cylinders. Three warnings had been recorded against it, and all three were right. The 2405.15848 template (an integrable structure whose defining operator has a kernel by construction) said that “commutation $\Rightarrow$ nonvanishing” is not a general principle; the Hammersley–Clifford obstruction said that “a multiplicative structure on $\{0, 1\}^n$ forces a product support” is false without positivity; and Terwilliger’s classification of the same theorem shape keeps nonvanishing and rigidity as *separate* axioms rather than deriving one from the other. The lesson is recorded in §9: the $e \leq 2$ proof closes not because of anything about ribbons but because at $e \leq 2$ the rank- d weight τ has at most two values and a square relation between them.

2 Setting and notation

We work throughout in the σ -coordinates of [1, Conv. 2.1]: for a shape weight W of rank e ,

$$\sigma(u) = (-1)^{|u|}W(u), \quad u \in \{0, 1\}^{e-1},$$

where $u = u_M(b, e)$ is the occupancy word of the bead move $b \mapsto b + e$, $u_i = [b + i \in M]$, and $|u| = \text{ht}$. Murnaghan–Nakayama is $\sigma \equiv \text{const}$. We freely use, and do not reprove, the following from [1]:

- [1, Lemma 3.1] (**lem:realise**): every window configuration may be prescribed freely by a partition. All the words appearing below are therefore realised by actual pairs of Maya diagrams, and no free parameter below is free for a trivial reason.
- [1, Lemma 4.1] (**lem:sectors**): the commutator has matrix elements only in the one-bead and two-bead sectors.
- [1, Lemma 4.2] (**lem:onebead**): with $X = M|_{(b, b+g)} = A m_e V m_f D$, $g = e + f$, $C = A m_e V$, $B = V m_f D$, the one-bead matrix element is, up to a global sign,

$$\sigma(D) \bar{\sigma}(C) - \sigma(A) \bar{\sigma}(B), \tag{1}$$

with $A, D \in \{0, 1\}^{e-1}$, $V \in \{0, 1\}^{f-e-1}$ and $m_e, m_f \in \{0, 1\}$ free and independent.

- [1, Lemma 4.3] (**lem:twobead**): for $2 \leq e$ and $1 \leq \delta \leq e - 1$ the two-bead matrix element is, up to a global sign,

$$\sigma(\pi 0 \pi') \bar{\sigma}(\pi' 0 \theta') - \sigma(\pi 1 \pi') \bar{\sigma}(\pi' 1 \theta'), \quad (2)$$

with $\pi \in \{0, 1\}^{\delta-1}$, $\pi' \in \{0, 1\}^{e-\delta-1}$, $\theta' \in \{0, 1\}^{\delta+f-e-1}$ free and independent; the marked letter sits at position δ of the σ -argument and at position $e - \delta$ of the $\bar{\sigma}$ -argument. For $e = 1$ the sector is empty.

Convention 2.1. Let σ be a shape weight of rank $m \geq 2$. Say that σ satisfies **(2B_m^{*})** if for every δ with $1 \leq \delta \leq m - 1$ and all $\pi, \varsigma \in \{0, 1\}^{\delta-1}$ and $\pi' \in \{0, 1\}^{m-\delta-1}$,

$$\sigma(\pi 0 \pi') \sigma(\pi' 0 \varsigma) = \sigma(\pi 1 \pi') \sigma(\pi' 1 \varsigma), \quad (3)$$

the marked letter sitting at position δ in the first factor and at position $m - \delta$ in the second. For $m = 1$ the condition is vacuous.

Say that σ satisfies the stronger **(2B_m^{**})** if the same identity holds with the second factor's prefix decoupled: for all $\alpha, \beta' \in \{0, 1\}^{\delta-1}$ and $\alpha', \beta \in \{0, 1\}^{m-\delta-1}$,

$$\sigma(\alpha 0 \alpha') \sigma(\beta 0 \beta') = \sigma(\alpha 1 \alpha') \sigma(\beta 1 \beta'). \quad (4)$$

Taking $\beta = \alpha'$, $\varsigma = \beta'$ shows **(2B_m^{**})** $\Rightarrow$ **(2B_m^{*})**.

It is convenient to record (3) in *window form*. Let $X = (X_1, \dots, X_{\delta+m-1}) \in \{0, 1\}^{\delta+m-1}$ and put $\Omega_c(X) = \sigma(X_{c+1}, \dots, X_{c+m-1})$. Then (3) at δ says:

$$\Omega_0(X) \Omega_\delta(X) \text{ is unchanged when } (X_\delta, X_m) = (0, 0) \text{ is replaced by } (1, 1), \quad (5)$$

all other letters of X being arbitrary and held fixed. (Indeed $\Omega_0 = \sigma(\pi X_\delta \pi')$ with $\pi = X_{[1, \delta-1]}$, $\pi' = X_{[\delta+1, m-1]}$, and $\Omega_\delta = \sigma(\pi' X_m \varsigma)$ with $\varsigma = X_{[m+1, \delta+m-1]}$, the letter X_m sitting at relative position $m - \delta$.)

3 Eliminating $\bar{\sigma}$: the one-bead sector is $\otimes$ -commutation

Proposition 3.1. Let $1 \leq e < f$ and let $\sigma, \bar{\sigma}$ be shape weights of ranks e, f over an integral domain K with fraction field F , both $\neq 0$. Then (1) vanishes for all admissible arguments if and only if there is a shape weight ρ of rank $f - e$ over F with

$$\bar{\sigma} = \sigma \otimes \rho = \rho \otimes \sigma. \quad (6)$$

In that case ρ is unique and $\rho \neq 0$.

Proof. Write $(1) = 0$ as

$$\sigma(D) \bar{\sigma}(A m_e V) = \sigma(A) \bar{\sigma}(V m_f D) \quad (A, D \in \{0, 1\}^{e-1}, V \in \{0, 1\}^{f-e-1}, m_e, m_f \in \{0, 1\}), \quad (7)$$

legitimate because $C = A m_e V$ and $B = V m_f D$ and all five blocks are free ([1, Lemma 3.1]).

($\Leftarrow$) If (6) holds then $\bar{\sigma}(A m_e V) = \sigma(A) \rho(V)$ by the first equality and $\bar{\sigma}(V m_f D) = \rho(V) \sigma(D)$ by the second, so both sides of (7) equal $\sigma(A) \sigma(D) \rho(V)$.

($\Rightarrow$) Fix once and for all a word $A_0 \in \{0, 1\}^{e-1}$ with $s_0 := \sigma(A_0) \neq 0$; it exists because $\sigma \neq 0$. Every cancellation below is by this one element s_0 . Specialising (7) at $A = A_0$ gives

$$\sigma(D) \bar{\sigma}(A_0 m_e V) = s_0 \bar{\sigma}(V m_f D), \quad (8)$$

and at $D = A_0$ gives

$$s_0 \bar{\sigma}(A m_e V) = \sigma(A) \bar{\sigma}(V m_f A_0). \quad (9)$$

Putting $D = A_0$ in (8) and cancelling s_0 (legitimate: K is a domain and $s_0 \neq 0$):

$$\bar{\sigma}(A_0 m_e V) = \bar{\sigma}(V m_f A_0) \quad \text{for all } m_e, m_f, V.$$

The left-hand side does not involve m_f and the right-hand side does not involve m_e ; since they are equal, both are independent of both letters. Define

$$\rho(V) := \bar{\sigma}(A_0 0 V) / s_0 \in F, \quad V \in \{0, 1\}^{f-e-1},$$

so that $\bar{\sigma}(A_0 m_e V) = s_0 \rho(V) = \bar{\sigma}(V m_f A_0)$ for either value of each marked letter. Substituting into (8) and cancelling s_0 gives $\bar{\sigma}(V m_f D) = \rho(V) \sigma(D)$, i.e. $\bar{\sigma} = \rho \otimes \sigma$; substituting into (9) and cancelling s_0 gives $\bar{\sigma}(A m_e V) = \sigma(A) \rho(V)$, i.e. $\bar{\sigma} = \sigma \otimes \rho$.

Uniqueness is forced by $\rho(V) = \bar{\sigma}(A_0 0 V) / s_0$. Finally $\rho \not\equiv 0$, since $\rho \equiv 0$ would give $\bar{\sigma} \equiv 0$. $\square$

Remark 3.2. Proposition 3.1 supersedes [1, Lemma 7.1] (`lem:flip`). The four flip relations (F0)–(F3) proved there are all immediate consequences of (6): (F0) is the fact that $\sigma \otimes \rho$ ignores the letter in position e and $\rho \otimes \sigma$ the one in position $f - e$; (F1)–(F3) are the two factorisations written out. More to the point, Proposition 3.1 assumes *no* nonvanishing.

Proposition 3.3. *Assume the situation of Proposition 3.1 and $e \geq 2$. Then (2) vanishes for all admissible arguments if and only if σ satisfies $(2B_e^*)$.*

Proof. Fix δ and split $\theta' = \varsigma c \psi$ with $|\varsigma| = \delta - 1$, $c \in \{0, 1\}$, $|\psi| = f - e - 1$; this is possible since $|\theta'| = \delta + f - e - 1 = (\delta - 1) + 1 + (f - e - 1)$. The word $\pi' b \theta' = \pi' b \varsigma c \psi$ of length $f - 1$ has first $e - 1$ letters $\pi' b \varsigma$, letter c in position e , and letters ψ in positions $e + 1, \dots, f - 1$. Hence $\bar{\sigma}(\pi' b \theta') = (\sigma \otimes \rho)(\pi' b \theta') = \sigma(\pi' b \varsigma) \rho(\psi)$, and (2) equals

$$[\sigma(\pi 0 \pi') \sigma(\pi' 0 \varsigma) - \sigma(\pi 1 \pi') \sigma(\pi' 1 \varsigma)] \cdot \rho(\psi).$$

If $(2B_e^*)$ holds this vanishes for every ψ . Conversely, choose ψ_0 with $\rho(\psi_0) \neq 0$ (possible as $\rho \not\equiv 0$) and cancel it; note this is a second, *separate* domain appeal, and its witness is $\rho(\psi_0)$, not s_0 . Since π, π', ς are free, (3) follows. $\square$

4 The Euclidean descent

Proof of Theorem 1.2. Induction on $m_1 + m_2$.

Base $m_1 = m_2 =: m$. Then $d = m$, and a word $y \in \{0, 1\}^{2m-1}$ has $y_{[1, m-1]} =: u$ and $y_{[m+1, 2m-1]} =: v$ free and independent, so the hypothesis reads $\sigma_1(u) \sigma_2(v) = \sigma_2(u) \sigma_1(v)$ for all u, v . Choose u_0 with $\sigma_1(u_0) \neq 0$ and set $\lambda = \sigma_2(u_0) / \sigma_1(u_0)$; taking $u = u_0$ gives $\sigma_2 = \lambda \sigma_1$, and $\lambda \neq 0$ as $\sigma_2 \neq 0$. Take $\tau = \sigma_1$, $c_1 = 1$, $c_2 = \lambda$.

Step. The hypothesis is symmetric in the two weights, so assume $m_1 < m_2$ and put $m_3 = m_2 - m_1 \geq 1$. Parametrise $y \in \{0, 1\}^{m_1+m_2-1}$ as

$$y = A \mu B \nu C, \quad |A| = |C| = m_1 - 1, \quad |B| = m_3 - 1, \quad \mu, \nu \in \{0, 1\},$$

which is a partition of the $m_1 + m_2 - 1$ letters because $2(m_1 - 1) + 2 + (m_3 - 1) = m_1 + m_2 - 1$. Then $y_{[1, m_1-1]} = A$, $y_{[m_1+1, m_1+m_2-1]} = B \nu C$, $y_{[1, m_2-1]} = A \mu B$ and $y_{[m_2+1, m_1+m_2-1]} = C$, so the hypothesis $\sigma_1 \otimes \sigma_2 = \sigma_2 \otimes \sigma_1$ reads

$$\sigma_1(A) \sigma_2(B \nu C) = \sigma_2(A \mu B) \sigma_1(C) \quad \text{for all } A, \mu, B, \nu, C. \quad (10)$$

Fix A_0 with $s_0 := \sigma_1(A_0) \neq 0$; again *every* cancellation below is by this one element.

- (i) Put $A = C = A_0$ in (10) and cancel s_0 : $\sigma_2(B \nu A_0) = \sigma_2(A_0 \mu B)$ for all B, μ, ν . The left side does not involve μ and the right side does not involve ν , so both are independent of both. Define $\sigma_3(B) := \sigma_2(A_0 0 B)/s_0 \in F$, a shape weight of rank m_3 ; then $\sigma_2(A_0 \mu B) = \sigma_2(B \nu A_0) = s_0 \sigma_3(B)$ for either marked letter.
- (ii) Put $C = A_0$ in (10): $\sigma_1(A) \sigma_2(B \nu A_0) = \sigma_2(A \mu B) s_0$. By (i) the left side is $\sigma_1(A) s_0 \sigma_3(B)$; cancel s_0 to get $\sigma_2(A \mu B) = \sigma_1(A) \sigma_3(B)$, i.e. $\sigma_2 = \sigma_1 \otimes \sigma_3$.
- (iii) Put $A = A_0$ in (10): $s_0 \sigma_2(B \nu C) = \sigma_2(A_0 \mu B) \sigma_1(C) = s_0 \sigma_3(B) \sigma_1(C)$; cancel s_0 to get $\sigma_2 = \sigma_3 \otimes \sigma_1$.

Hence $\sigma_1 \otimes \sigma_3 = \sigma_2 = \sigma_3 \otimes \sigma_1$, and $\sigma_3 \neq 0$ because $\sigma_2 \neq 0$. Since $\gcd(m_1, m_3) = \gcd(m_1, m_2) = d$ and $m_1 + m_3 < m_1 + m_2$, the inductive hypothesis gives a nonzero rank- d weight τ and $c_1, c_3 \in F^\times$ with $\sigma_1 = c_1 \tau^{\otimes(m_1/d)}$ and $\sigma_3 = c_3 \tau^{\otimes(m_3/d)}$. By (ii) and associativity of $\otimes$,

$$\sigma_2 = \sigma_1 \otimes \sigma_3 = c_1 c_3 \tau^{\otimes(m_1/d)} \otimes \tau^{\otimes(m_3/d)} = c_1 c_3 \tau^{\otimes(m_2/d)}. \quad \square$$

Remark 4.1 (what the descent is). Steps (i)–(iii) are the standard proof that two commuting words in a free monoid are powers of a common word, transported to the graded semigroup of Convention 1.1. The statement “ $\sigma_2 = \sigma_1 \otimes \sigma_3 = \sigma_3 \otimes \sigma_1$ ” is the assertion that σ_1 is both a prefix and a suffix of σ_2 and that the two overlapping occurrences agree — the defect theorem. No positivity, no nonvanishing, and no invertibility of any weight value is used: the single witness s_0 does all the work.

5 The two-bead relation descends to rank d

Lemma 5.1. *Let $d \mid m$, $k = m/d$, let τ be a nonzero shape weight of rank d over a domain, and put $\sigma = \tau^{\otimes k}$ (rank m).*

- (a) *If τ satisfies $(2B_d^{**})$ then σ satisfies $(2B_m^*)$. (For $k = 1$ this is trivial, and $(2B_d^*)$ suffices.)*
- (b) *If σ satisfies $(2B_m^*)$ then τ satisfies $(2B_d^*)$.*

Proof. Work in the window form (5). For $X \in \{0, 1\}^{\delta+m-1}$ put $T_c := \tau(X_{c+1}, \dots, X_{c+d-1})$. Unwinding Convention 1.1,

$$\Omega_c(X) = \sigma(X_{c+1}, \dots, X_{c+m-1}) = \prod_{i=0}^{k-1} T_{c+id},$$

the sites $c + d, c + 2d, \dots, c + (k - 1)d$ being the ignored letters. So

$$\Omega_0(X) \Omega_\delta(X) = \prod_{i=0}^{k-1} T_{id} \cdot \prod_{i=0}^{k-1} T_{\delta+id}.$$

The factor T_c involves the site s iff $c < s < c + d$. Now:

- *site δ .* In the first product $c = id$, so we need $id < \delta < (i + 1)d$: this happens iff $d \nmid \delta$, and then for the single index $p := \lfloor \delta/d \rfloor$. In the second product $c = \delta + id \geq \delta$, so never.
- *site $m = kd$.* In the first product we would need $id < kd < (i + 1)d$ with $i \leq k - 1$: impossible. In the second we need $i < (m - \delta)/d < i + 1$, which happens iff $d \nmid \delta$, and then for the single index $i = k - p - 1$; the corresponding factor is $T_{\delta+(k-p-1)d} = T_{(k-1)d+\delta_0}$, where $\delta_0 := \delta - pd \in [1, d - 1]$.

Case $d \mid \delta$. No factor involves X_δ or X_m , so (5) holds identically; nothing to prove in either direction.

Case $d \nmid \delta$, $\delta = pd + \delta_0$ with $1 \leq \delta_0 \leq d - 1$. Let Π denote the product of the remaining $2k - 2$ factors; Π does not involve X_δ or X_m . Writing $T^{(b)}$ for the value with $(X_\delta, X_m) = (b, b)$, (5) at δ reads

$$\Pi \cdot T_{pd}^{(0)} T_{(k-1)d+\delta_0}^{(0)} = \Pi \cdot T_{pd}^{(1)} T_{(k-1)d+\delta_0}^{(1)}. \quad (11)$$

Explicitly, $T_{pd}^{(b)} = \tau(\alpha b \alpha')$ with $\alpha = X_{[pd+1, \delta-1]}$ of length $\delta_0 - 1$ and $\alpha' = X_{[\delta+1, pd+d-1]}$ of length $d - \delta_0 - 1$, the marked letter at position δ_0 ; and $T_{(k-1)d+\delta_0}^{(b)} = \tau(\beta b \beta')$ with $\beta = X_{[(k-1)d+\delta_0+1, m-1]}$ of length $d - \delta_0 - 1$ and $\beta' = X_{[m+1, (k-1)d+\delta_0+d-1]}$ of length $\delta_0 - 1$, the marked letter at position $d - \delta_0$.

(a) If τ satisfies $(2B_d^{**})$ then $T^{(0)}T^{(0)} = T^{(1)}T^{(1)}$ in (11) — the four blocks $\alpha, \alpha', \beta, \beta'$ have exactly the lengths required by (4) — so (11) holds whatever Π is. Since δ was arbitrary, σ satisfies $(2B_m^*)$.

(b) Fix $\delta_0 \in [1, d - 1]$ and take $p = k - 1$, i.e. $\delta = (k - 1)d + \delta_0$; this is legitimate since $\delta \leq (k - 1)d + d - 1 = m - 1$. Then the two marked windows are $((k - 1)d, kd)$ and $((k - 1)d + \delta_0, (k - 1)d + \delta_0 + d)$, which overlap in $((k - 1)d + \delta_0, kd)$; comparing the displays above, $\beta = \alpha'$, so (11) is an instance of (3) multiplied by Π . The remaining factors are T_{id} for $0 \leq i \leq k - 2$, supported on the disjoint windows $(id, (i + 1)d) \subseteq (0, (k - 1)d)$, and $T_{\delta+id}$ for $1 \leq i \leq k - 1$, supported on the disjoint windows $(\delta + id, \delta + (i + 1)d) \subseteq (kd + \delta_0, \delta + m)$. All $2k - 2$ of these windows are pairwise disjoint and disjoint from $((k - 1)d, kd + \delta_0)$, which carries $\alpha, \alpha' = \beta, \beta'$ and the marked sites. Choose u^+ with $\tau(u^+) \neq 0$ and set X equal to u^+ on each of the $2k - 2$ windows; then $\Pi = \tau(u^+)^{2k-2} \neq 0$ because the ring is a domain. Cancelling this one named element from (11) gives (3) at δ_0 for τ , with $\alpha, \alpha' = \beta, \beta'$ free. As δ_0 was arbitrary, τ satisfies $(2B_d^*)$. $\square$

Remark 5.2 (both implications of Lemma 5.1 are strict). Part (b) yields only $(2B_d^*)$, not $(2B_d^{**})$, because for $p \leq k - 2$ the block β sits inside one of the Π -windows, so “ $\Pi \neq 0$ ” and “ β arbitrary” cannot be arranged at once. That obstruction is real, not an artefact of the proof: *neither implication of Lemma 5.1 can be reversed.*

For $d \leq 3$ the conditions $(2B_d^*)$ and $(2B_d^{**})$ have the same solutions, but for $d \geq 4$ the second is strictly stronger (Check V5': at $d = 4$ they admit 6 resp. 2 proper supports, at $d = 5$, 24 resp. 8). And at $d = 4$, $k = 2$ there are exactly four weights over $\mathbb{F}_3$, up to scaling, separating the middle condition from the strong one (Check V5): the singletons

$$\tau = \mathbb{K}_w, \quad w \in \{100, 110, 001, 011\}.$$

Each satisfies $(2B_4^*)$ but not $(2B_4^{**})$, and $\tau^{\otimes 2}$ nevertheless satisfies $(2B_8^*)$ — as does $\tau^{\otimes 3}$ against $(2B_{12}^*)$, all 2 096 128 instances. So for $d \geq 4$ and $k \geq 2$,

$$\{\tau \text{ with } (2B_d^{**})\} \subsetneq \{\tau : \tau^{\otimes k} \text{ has } (2B_{kd}^*)\} \subsetneq \{\tau \text{ with } (2B_d^*)\},$$

both inclusions strict. Consequently Theorem 6.6, which routes through $(2B_d^{**})$, produces *some* but not all commuting pairs with zeros — the four singletons give further ones at $e = 8$ — and Theorem 1.3 covers those because it is stated with $(2B_e^*)$ on σ , exactly what the commutator delivers. Nothing in Theorem 1.3, Corollary 1.4, Theorem 6.1 or Theorem 6.6 depends on identifying the middle condition; see §8, (H1).

6 The classification and the corrected nonvanishing statement

Proof of Theorem 1.3. By [1, Lemma 4.1] the commutator vanishes iff (1) and (2) vanish for all admissible arguments.

($\Rightarrow$) By Proposition 3.1 there is a nonzero rank- $(f-e)$ weight ρ over F with $\sigma \otimes \rho = \bar{\sigma} = \rho \otimes \sigma$. As $\gcd(e, f-e) = \gcd(e, f) = d$, Theorem 1.2 gives a nonzero rank- d weight τ over F and $c, c' \in F^\times$ with $\sigma = c\tau^{\otimes(e/d)}$ and $\rho = c'\tau^{\otimes((f-e)/d)}$; then $\bar{\sigma} = \sigma \otimes \rho = cc'\tau^{\otimes(f/d)}$. If $e \geq 2$, Proposition 3.3 gives $(2B_e^*)$ for σ ; if $e = 1$ there is nothing to prove.

($\Leftarrow$) Given the stated form, set $\rho := \bar{c}c^{-1}\tau^{\otimes((f-e)/d)}$. Associativity of $\otimes$ gives $\sigma \otimes \rho = \bar{c}\tau^{\otimes(f/d)} = \bar{\sigma} = \rho \otimes \sigma$, so (1) vanishes by Proposition 3.1; and (2) vanishes by Proposition 3.3 since σ satisfies $(2B_e^*)$. $\square$

Proof of Corollary 1.4. $\sigma(u) = c \prod_{i=1}^{e/d} \tau(u_{[(i-1)d+1, id-1]})$ and similarly for $\bar{\sigma}$. If τ vanishes nowhere then, F being a field, neither does σ or $\bar{\sigma}$. Conversely if $\tau(w) = 0$ for some $w \in \{0, 1\}^{d-1}$ then any u whose first $d-1$ letters are w has $\sigma(u) = 0$, and likewise for $\bar{\sigma}$. (For $d = e$ the first statement needs $e/d = 1$; the argument is the same.) Finally $W = \pm\sigma$ and $\bar{W} = \pm\bar{\sigma}$ pointwise. The two cases $\gcd \leq 2$ and $\gcd \geq 3$ are Theorem 6.1 and Theorem 6.6 below. $\square$

6.1 $\gcd(e, f) \leq 2$: nonvanishing holds

Theorem 6.1. *Let $1 \leq e < f$ with $d = \gcd(e, f) \leq 2$, let K be an integral domain, and let $W \not\equiv 0 \not\equiv \bar{W}$ be shape weights of ranks e, f over K with $[R_e^W, R_f^{\bar{W}}] = 0$. Then no value of W or of $\bar{W}$ is zero.*

Proof. By Theorem 1.3 write $\sigma = c\tau^{\otimes(e/d)}$, $\bar{\sigma} = \bar{c}\tau^{\otimes(f/d)}$ with $\tau \not\equiv 0$ of rank d . By Corollary 1.4 it is enough to show τ vanishes nowhere.

If $d = 1$ then τ is a shape weight of rank 1, i.e. a single scalar t , and $\tau \not\equiv 0$ means $t \neq 0$. There is nothing to vanish. (Note that this case uses *only* the one-bead sector.)

If $d = 2$ then $\tau : \{0, 1\} \rightarrow F$ and $d \mid e$ forces $e \geq 2$, so the two-bead sector is nonempty and Proposition 3.3 applies: σ satisfies $(2B_e^*)$. By Lemma 5.1(b), τ satisfies $(2B_2^*)$. The only instance of (3) at $m = 2$ is $\delta = 1$ with π, π', ς all empty, namely

$$\tau(0)^2 = \tau(1)^2.$$

If $\tau(0) = 0$ then $\tau(1)^2 = 0$, hence $\tau(1) = 0$ because F is a field, hence $\tau \equiv 0$ — a contradiction. Symmetrically $\tau(1) \neq 0$. $\square$

Corollary 6.2 (the coprime headline, unconditionally). *Let $\gcd(e, f) = 1$ and $[R_e^W, R_f^{\bar{W}}] = 0$ with $W \not\equiv 0 \not\equiv \bar{W}$ over an integral domain. Then*

$$W(u) = \alpha(-1)^{|u|}, \quad \bar{W}(y) = \beta(-1)^{|y|}$$

for some $\alpha, \beta \neq 0$; that is, $R_e^W = \alpha p_e \cdot$ and $R_f^{\bar{W}} = \beta p_f \cdot$. In particular, for $(e, f) = (1, 3)$ the cross-rank commutator forces $\bar{W}_{(2,1)} = \bar{W}_{(1,2)}$ and then the full alternation.

Proof. $d = 1$, so τ is a scalar $t \neq 0$ and $\tau^{\otimes k}$ is the constant function t^k ; hence σ and $\bar{\sigma}$ are nonzero constants, which is the assertion. This is [1, Cor. 1.5], now with no appeal to [1, Lemma 7.3] — and, remarkably, with no appeal to the two-bead sector either. $\square$

Remark 6.3. Corollary 6.2 also reproves [1, Prop. 7.7] (**prop:e1**, the case $e = 1$) in one line, and it does so in the regime where the two-bead sector is empty, which is exactly the regime **prop:e1** was written to handle by hand.

6.2 $\gcd(e, f) \geq 3$: nonvanishing fails

Convention 6.4. For $d \geq 3$, $1 \leq \delta \leq d-1$ with $2\delta \neq d$, and $b \in \{0, 1\}$ put

$$\mathcal{C}_{\delta,b} := \{w \in \{0, 1\}^{d-1} : w_\delta = b, w_{d-\delta} = 1-b\},$$

a nonempty set of size 2^{d-3} (it is nonempty precisely because $\delta \neq d-\delta$), and let $\tau_{\delta,b}$ be its indicator function, a shape weight of rank d .

Lemma 6.5. $\tau_{\delta,b}$ satisfies $(2B_d^{**})$, hence also $(2B_d^*)$.

Proof. Write $\tau = \tau_{\delta,b}$, $\mathcal{C} = \mathcal{C}_{\delta,b}$. Since τ is $\{0, 1\}$ -valued, (4) at δ_0 says

$$[u^0 \in \mathcal{C} \text{ and } v^0 \in \mathcal{C}] \iff [u^1 \in \mathcal{C} \text{ and } v^1 \in \mathcal{C}],$$

where $u^c = \alpha c \alpha'$ carries c in position δ_0 and $v^c = \beta c \beta'$ carries c in position $d - \delta_0$, the other letters being fixed. Three cases.

If $\delta_0 = \delta$: then $u^c \in \mathcal{C}$ requires $u_\delta^c = b$, i.e. $c = b$; and v^c carries c in position $d - \delta_0 = d - \delta$, so $v^c \in \mathcal{C}$ requires $c = 1 - b$. The two demands are incompatible, so both sides of the equivalence are false.

If $\delta_0 = d - \delta$: symmetrically, $u^c \in \mathcal{C}$ requires $c = 1 - b$ (position $d - \delta$) and $v^c \in \mathcal{C}$ requires $c = b$ (position $d - \delta_0 = \delta$); again both sides are false.

If $\delta_0 \notin \{\delta, d - \delta\}$: the marked position of u^c is $\delta_0 \neq \delta, d - \delta$, so membership $u^c \in \mathcal{C}$ is decided by the fixed letters and does not depend on c ; and the marked position of v^c is $d - \delta_0$, which differs from δ (else $\delta_0 = d - \delta$) and from $d - \delta$ (else $\delta_0 = \delta$), so $v^c \in \mathcal{C}$ likewise does not depend on c . Both sides are then equal. $\square$

Theorem 6.6. Let $1 \leq e < f$ with $d = \gcd(e, f) \geq 3$, let K be any integral domain, and let $\tau = \tau_{1,b}$ of Convention 6.4. Put

$$\sigma := \tau^{\otimes(e/d)}, \quad \bar{\sigma} := \tau^{\otimes(f/d)}, \quad W(u) := (-1)^{|u|} \sigma(u), \quad \bar{W}(y) := (-1)^{|y|} \bar{\sigma}(y).$$

Then $W \neq 0 \neq \bar{W}$ and $[R_e^W, R_f^{\bar{W}}] = 0$, while W vanishes at every $u \in \{0, 1\}^{e-1}$ one of whose blocks $u_{[(i-1)d+1, id-1]}$ lies outside $\mathcal{C}_{1,b}$ — that is, at $2^{e-1} - 2^{(e/d)(d-3)} \cdot 2^{(e/d)-1}$ of the 2^{e-1} words, a positive number; likewise for $\bar{W}$. In particular [1, Lemma 7.3] is false for every such (e, f) , and with it [1, Theorem 1.4] (necessity). This is a sufficiency statement only: it exhibits counterexamples, and by Remark 5.2 not all of them.

Proof. $\tau \neq 0$ since $\mathcal{C}_{1,b} \neq \emptyset$ (here $d \geq 3$ gives $1 \neq d-1$), so σ and $\bar{\sigma}$ are not identically zero. By Lemma 6.5 τ satisfies $(2B_d^{**})$, so by Lemma 5.1(a) σ satisfies $(2B_e^*)$. By Theorem 1.3 (direction $\Leftarrow$, with $c = \bar{c} = 1$) the commutator vanishes. Finally $\mathcal{C}_{1,b} \neq \{0, 1\}^{d-1}$, because $d \geq 3$ means the constraint $w_1 = b, w_{d-1} = 1 - b$ excludes e.g. the word w with $w_1 = w_{d-1} = b$; so τ has a zero, and Corollary 1.4 transports it to W and $\bar{W}$.

That this contradicts [1, Lemma 7.3] is immediate. It contradicts [1, Thm. 1.4] (necessity) as well: case (c) there asserts $W = \alpha W_e^\gamma$ with γ valued in $F^\times$, and every such W is nowhere zero, while (a) and (b) are excluded by $W \neq 0 \neq \bar{W}$. $\square$

Theorem 6.7 (the smallest case, explicitly). Let K be any integral domain and let $W, \bar{W}$ be the shape weights of ranks 3 and 6 given in the ribbon-shape indexing of [1, Lemma 2.2] by

$$\begin{aligned} W_{(2,1)} &= -1, & W_{(3)} &= W_{(1,1,1)} = 0; \\ \bar{W}_{(2,3,1)} &= +1, & \bar{W}_{(2,1,2,1)} &= -1, & \bar{W}_\alpha &= 0 \text{ for the other 30 compositions } \alpha \models 6. \end{aligned}$$

Then $[R_3^W, R_6^{\bar{W}}] = 0$, and both operators are nonzero.

Proof. This is Theorem 6.6 with $d = e = 3$, $f = 6$, $b = 1$: then $\mathcal{C}_{1,1} = \{w \in \{0,1\}^2 : w_1 = 1, w_2 = 0\} = \{(1,0)\}$, so $\tau = \sigma$ is the indicator of the occupancy word $(1,0)$, whose composition is $\alpha(1,0) = (3-1, 1-0) = (2,1)$ by [1, Lemma 2.2]; and $W(1,0) = (-1)^1 \cdot 1 = -1$. For $\bar{\sigma} = \tau^{\otimes 2}$ the support is $\{y \in \{0,1\}^5 : (y_1, y_2) = (y_4, y_5) = (1,0)\} = \{(1,0,0,1,0), (1,0,1,1,0)\}$, with $\bar{\sigma} = 1$ there; the corresponding compositions are $(2,3,1)$ and $(2,1,2,1)$, and $\bar{W} = (-1)^{|y|} \bar{\sigma}$ gives $+1$ and -1 . Nonvanishing of the operators is [1, Lemma 3.1]: both ribbon shapes occur. $\square$

Remark 6.8 (why this one commutes, informally). R_3^W performs the local move $1100 \mapsto 0101$ on the Maya diagram: it needs $b, b+1 \in M$ and $b+2, b+3 \notin M$. The support of $\bar{W}$ is exactly the set of configurations from which the *same bead* can travel $b \mapsto b+3 \mapsto b+6$, in one of the two possible orders — the two orders being separated by the occupancy of the site $b+3$, which is the free letter y_3 . So $R_6^{\bar{W}}$ is the “one-bead part of $(R_3^W)^2$ ”, up to the sign bookkeeping, and $\otimes$ is the operation that builds it. Theorem 1.2 says this is the *only* way a commuting partner can arise.

6.3 Where the old classification was right

Proposition 6.9. *Let τ be a nowhere-zero shape weight of rank m over a field F satisfying $(2B_m^*)$. Then there are $\gamma_1, \dots, \gamma_{m-1} \in F^\times$ with*

$$\tau(w) = \tau(0^{m-1}) \prod_{j=1}^{m-1} \gamma_j^{w_j}, \quad \gamma_\delta \gamma_{m-\delta} = 1 \quad (1 \leq \delta \leq m-1).$$

Proof. For $1 \leq j \leq m-1$ let $\theta_j(w) := \tau(w | w_j = 1) / \tau(w | w_j = 0) \in F^\times$, a function of the letters w_i , $i \neq j$. Dividing (3) at δ by its left-hand side gives

$$\theta_\delta(\pi \cdot \pi') \cdot \theta_{m-\delta}(\pi' \cdot \varsigma) = 1,$$

where the first factor is a function of (π, π') only and the second of (π', ς) only. Fix π' and vary π : the second factor cannot change, so the first is constant in π ; varying ς shows the second is constant in ς . Hence $\theta_\delta(w)$ depends only on $w_{[\delta+1, m-1]}$, and $\theta_{m-\delta}(w)$ depends only on $w_{[1, m-\delta-1]}$. Replacing δ by $m-\delta$ in the latter (also in $[1, m-1]$) gives: $\theta_\delta(w)$ depends only on $w_{[1, \delta-1]}$. The two index sets $[\delta+1, m-1]$ and $[1, \delta-1]$ are disjoint, so θ_δ is a constant γ_δ . Then (3) reads $\gamma_\delta \gamma_{m-\delta} = 1$, and flipping the letters of 0^{m-1} one at a time gives the product formula. $\square$

Corollary 6.10. *In Theorem 1.3, the solutions with τ nowhere zero are exactly [1, Thm. 1.4(c)]: $\sigma = \alpha \sigma_e^\gamma$, $\bar{\sigma} = \beta \sigma_f^\gamma$ with $\gamma : \mathbb{Z}/d \rightarrow F^\times$, $\gamma(0) = 1$, $\gamma(\delta)\gamma(-\delta) = 1$.*

Proof. By Lemma 5.1(b), τ satisfies $(2B_d^*)$; being nowhere zero, Proposition 6.9 applies at $m = d$ and gives $\tau(w) = \tau(0^{d-1}) \prod_j \gamma_j^{w_j}$ with $\gamma_\delta \gamma_{d-\delta} = 1$. Then

$$\sigma(u) = c \prod_{i=1}^{e/d} \tau(u_{[(i-1)d+1, id-1]}) = \sigma(0^{e-1}) \prod_{k=1}^{e-1} \gamma_{k \bmod d}^{u_k},$$

with the convention $\gamma_0 := 1$ (the letters in positions $d, 2d, \dots$ are the ones $\otimes$ ignores), and likewise for $\bar{\sigma}$. This is exactly [1, Conv. 5.1]. Conversely every such pair has nowhere-zero τ . $\square$

So [1, Thm. 1.4] is not wrong about the solutions it lists; it is incomplete. Its list is the nowhere-zero locus, which is everything precisely when $d \leq 2$.

7 Verification

All code is at `proofs/code/2026-09-16-c1-Q150/`. The numeric commutator checker `direct.py` is written from [1, Conv. 2.2] and shares no algebra with `solver.py` of [1].

V0 (calibration). On $(e, f) = (2, 3)$, `direct.py` and `solver.commutator_eqs_window` agree on all 300 random weight pairs over $\mathbb{F}_3$; the population is live, 16 commuting and 284 not. A first version of `direct.py` silently dropped the legal moves originating below the frozen tail; it was corrected before use. A first run compared residuals over $\mathbb{Z}$ against solutions over $\mathbb{F}_3$ and reported four false failures (residuals $\pm 3, \pm 6$); the checker now takes a modulus.

V0' (the counterexample, two code paths). For Theorem 6.7: `direct.py` finds 0 nonzero commutator matrix elements; `solver.commutator_eqs_window(3,6,Lw=14)` with the same numeric weights sweeps 16384 states, 4192 matrix elements, and returns 0 nonzero residuals over $\mathbb{Z}$. Positive controls at $(3, 6)$: Murnaghan–Nakayama commutes; the γ -character with $d = 3$, $\gamma(1) = 5$, $\gamma(2) = 1/5$ commutes. Negative control: perturbing one value of that $\bar{W}$ by 1 makes the commutator nonzero.

V1 (structure theorem). For $(e, f) \in \{(2, 3), (2, 4), (3, 4), (3, 5), (3, 6), (4, 5), (4, 6)\}$, every solution of the reduced system over $\mathbb{F}_3$ — 4, 8, 4, 4, 16, 4, 8 solutions respectively — is of the form $\sigma = c\tau^{\otimes(e/d)}$, $\rho = c'\tau^{\otimes((f-e)/d)}$ for a rank- d τ . Zero exceptions.

V2 (counterexamples). For $(e, f) \in \{(3, 6), (3, 9), (6, 9), (4, 8), (5, 10), (6, 12), (9, 12), (6, 15)\}$ and $\tau = \tau_{1,b}$: 0 violations of $(2B_e^*)$ and 0 of (6); and for $e + f \leq 15$ the direct commutator is zero. The zero counts are recorded and factor as predicted, e.g. at $(6, 9)$: W has 30 zeros out of 32 ($2 = 2^{6-1-4}$ survivors, $\mathcal{C}_{1,1}$ having size $2^{3-3} = 1$ per block and two blocks), $\bar{W}$ has 252 out of 256.

V3 (gcd ≤ 2). Exhaustively over $\mathbb{F}_5$ for $(e, f) \in \{(2, 3), (2, 4), (2, 5), (3, 4), (3, 5), (3, 7), (4, 5), (4, 6), (4, 7), (5, 6), (4, 8)\}$, no solution has a vanishing weight value.

V4 (negative controls, all must fire). (a) lifting a rank-3 τ to $(e, f) = (6, 10)$, where $3 \nmid 10$, violates (6); (b) $\tau = \mathbb{K}_{(0,0)}$, which fails $(2B_3^*)$, lifted to $(6, 9)$ violates $(2B_6^*)$; (c) the direct commutator for (b) is nonzero. All three fire.

V5 (descent). Over $\mathbb{F}_3$, all nonzero rank- d weights τ , for $d \in \{2, 3, 4\}$ and $k \in \{1, 2, 3\}$ with $2^{kd-1} \leq 256$. At $k = 1$ the middle condition is $(2B_d^*)$ itself (8/8, 80/80, 6560/6560 weights at $d = 2, 3, 4$). At $k \geq 2$, $(2B_d^{**})$ implies $(2B_{kd}^*)$ with no exception, as Lemma 5.1(a) requires; the converse holds at $d = 2$ (8/8) and $d = 3$ (80/80) but **fails at** $d = 4$: 6552 of 6560 weights agree, and the eight that do not are the four singletons $\mathbb{K}_{100}, \mathbb{K}_{110}, \mathbb{K}_{001}, \mathbb{K}_{011}$ and their doubles, which satisfy $(2B_8^*)$ after lifting while failing $(2B_4^{**})$. A separate run confirms the same four satisfy $(2B_{12}^*)$ at $k = 3$ (all 2096 128 instances). This is Remark 5.2, and I record how it surfaced: the check was written expecting agreement, and its assertion caught a claim that had already been written into the draft.

V5' (the two two-bead conditions differ). Enumerating all admissible supports: at $d = 2$ both conditions admit only the full support; at $d = 3$ both admit the two proper supports $\{(0, 1)\}, \{(1, 0)\}$; at $d = 4$, $(2B^*)$ admits 6 proper supports but $(2B^{**})$ only 2; at $d = 5$, 24 against 8. This is the content of Remark 5.2.

Scope, stated against the quantifier. The exhaustive scans V1 and V3 range over $\mathbb{F}_p$ for $p \in \{3, 5\}$ and over the pairs listed; they are not a proof for other (e, f) or other characteristics, and they are not needed for one — §§3–6 are unconditional. One earlier scan of mine *was* too narrow and gave a false negative: searching only *singleton*-supported σ reported “no zero-bearing solution” at $(6, 9)$, $(9, 12)$, $(8, 12)$, $(10, 15)$ — all with $\gcd \geq 3$ — and suggested the criterion “ $e \mid f$ ”. The composite $\tau^{\otimes 2}$ at $(6, 9)$ refutes that criterion. The reported counts had been 12, 104, 48, 192 candidate words passing $(2B^*)$; the quantifier those counts answered was “how many *singletons*”, not “how many weights”.

8 What is and is not proved

Proved

1. Proposition 3.1: the one-bead sector is exactly $\bar{\sigma} = \sigma \otimes \rho = \rho \otimes \sigma$. Unconditional; no nonvanishing.
2. Proposition 3.3: given that, the two-bead sector is exactly $(2B_e^*)$ for σ alone. $\bar{\sigma}$ is eliminated entirely.
3. Theorem 1.2: commuting weights are powers of a common rank-gcd weight. Unconditional.
4. Lemma 5.1: $(2B_d^{**}) \Rightarrow (2B_m^*) \Rightarrow (2B_d^*)$ for $\sigma = \tau^{\otimes(m/d)}$.
5. Theorem 1.3 and Corollary 1.4: the classification and the reduction of nonvanishing to nonvanishing of τ .
6. Theorem 6.1: nonvanishing for $\gcd(e, f) \leq 2$, hence Corollary 6.2, hence [1, Cor. 1.5, Prop. 7.7] unconditionally.
7. Theorem 6.6, Theorem 6.7, Lemma 6.5: explicit counterexamples for every $\gcd(e, f) \geq 3$.
8. Corollary 6.10: the nowhere-zero part of the new classification is exactly [1, Thm. 1.4(c)].

Not proved

(H1) **The exact rank- d form of $(2B_e^*)$ when $e > d$.** I prove $(2B_d^{**}) \Rightarrow (2B_e^*) \Rightarrow (2B_d^*)$, and Remark 5.2 shows *both* implications are strict for $d \geq 4$: the condition in the middle is a genuinely new one, sitting strictly between, and I do not have a closed form for it. The four singletons at $d = 4$ say what a proof must accommodate — “for every β there is η with $\tau(\eta\beta) \neq 0$ ” is exactly what a singleton τ violates, yet its lift still satisfies $(2B_{kd}^*)$ — so the cofactor-cancellation route cannot be rescued by a better choice of witness. Whether the middle condition depends on k I do not know; it does not for $k \in \{2, 3\}$ at $d \leq 4$. *This does not affect any theorem above*: the classification is stated with $(2B_e^*)$ on σ , which is exactly what the commutator gives.

- (H2) **A classification of the rank- d solutions of $(2B_d^*)$ with zeros.** Computationally (V5') the admissible supports at $d \leq 5$ are the full support and the subsets of the $\mathcal{C}_{\delta,b}$; I prove only that each $\mathcal{C}_{\delta,b}$ is admissible (Lemma 6.5), not that every proper admissible support is contained in one, nor which value assignments on such a support satisfy $(2B_d^*)$. This is now the interesting question: it is a finite, rank- d -only question, and it is what a complete description of the $\gcd \geq 3$ locus requires.
- (H3) **Shared weights and more than two ranks.** Gaps (G3), (G4) of [1] are untouched. Note that (G4) looks easier now: requiring $[R_{e_i}^{W_i}, R_{e_j}^{W_j}] = 0$ for all pairs from a set makes all the W_i powers of a common τ pairwise, and the question is whether the τ 's can be taken equal — again a commuting-words question.
- (H4) **(G2) of [1]**, the $\mathbb{Z}/2$ at $\delta = d/2$ being a gauge, is unaffected and still open.

9 Post-mortem: why $e \leq 2$ closed and $e \geq 3$ could not

The brief asked for the step in the $e \leq 2$ proof that uses ribbon structure, on the grounds that the counterexample template of 2405.15848 shows that commutation alone cannot force nonvanishing. The answer is that there is no such step. The $e \leq 2$ proof closes for a reason that has nothing to do with ribbons and everything to do with arithmetic: at $e \leq 2$ one has $d = \gcd(e, f) \leq 2$, and a rank- ≤ 2 weight τ has at most two values tied by $\tau(0)^2 = \tau(1)^2$, so a zero in one forces a zero in the other. At $e = 3$ with $3 \nmid f$ the lemma is still true, but again for the arithmetic reason $d \leq 2$; and at $e = 3$, $f = 6$ it is false. The checks C5 and C6 of [1] covered $(1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (3, 4), (3, 5)$: every one of these has $\gcd \leq 2$. The smallest pair with $\gcd \geq 3$ is $(3, 6)$, and it was never run.

Three further notes, recorded because they are the transferable part.

- *The cylinder bookkeeping was the wrong object.* Gap (G1) described the obstruction as “the cylinder produced has length $e - \delta$, not $e - 1$, so the induction does not terminate in one step”. That is a true description of a route to a false theorem. The reason the induction would not terminate is that there was nothing at the end of it.
- *Eliminating the larger unknown first.* Every attempt so far reasoned about the pair $(\sigma, \bar{\sigma})$, with $2^{e-1} + 2^{f-1}$ unknowns. Proposition 3.1 replaces $\bar{\sigma}$ by ρ of rank $f - e$, and then the two-bead sector becomes a statement about σ alone. The classification is then visibly a question about a *semigroup*, and the semigroup has a classical theorem.
- *A negative scan must be audited against its quantifier.* See the last paragraph of §7. A scan over singleton supports reported “no counterexample at $(6, 9)$ ” and would have produced the wrong theorem (“ $e \mid f$ ” instead of “ $\gcd(e, f) \geq 3$ ”). The composite $\tau^{\otimes 2}$ is not a singleton.

References

- [1] Clio, *Shape weights on ribbons: the commuting locus is the transpose-equivariant $\gcd(e, f)$ -periodic characters* (Q149), `proofs/2026-09-15-c1-Q149-shape-weights.tex`, 15 September 2026. Numbering used above: Conv. 2.1 = `conv:sigma`, Conv. 2.2 = `conv:def`, Lemma 2.2 = `lem:dict`, Lemma 3.1 = `lem:realise`, Lemma 4.1 = `lem:sectors`, Lemma 4.2 = `lem:onebead`, Lemma 4.3 = `lem:twobead`, Thm. 1.4 = `thm:main`, Cor. 1.5 = `cor:coprime`, Conv. 5.1 = `conv:gamma`, Lemma 7.1 = `lem:flip`, Lemma 7.3 = `lem:nonvanish`, Prop. 7.7 = `prop:e1`.
- [2] Clio, *Ribbon commutators on Maya diagrams* (Q92), September 2026.